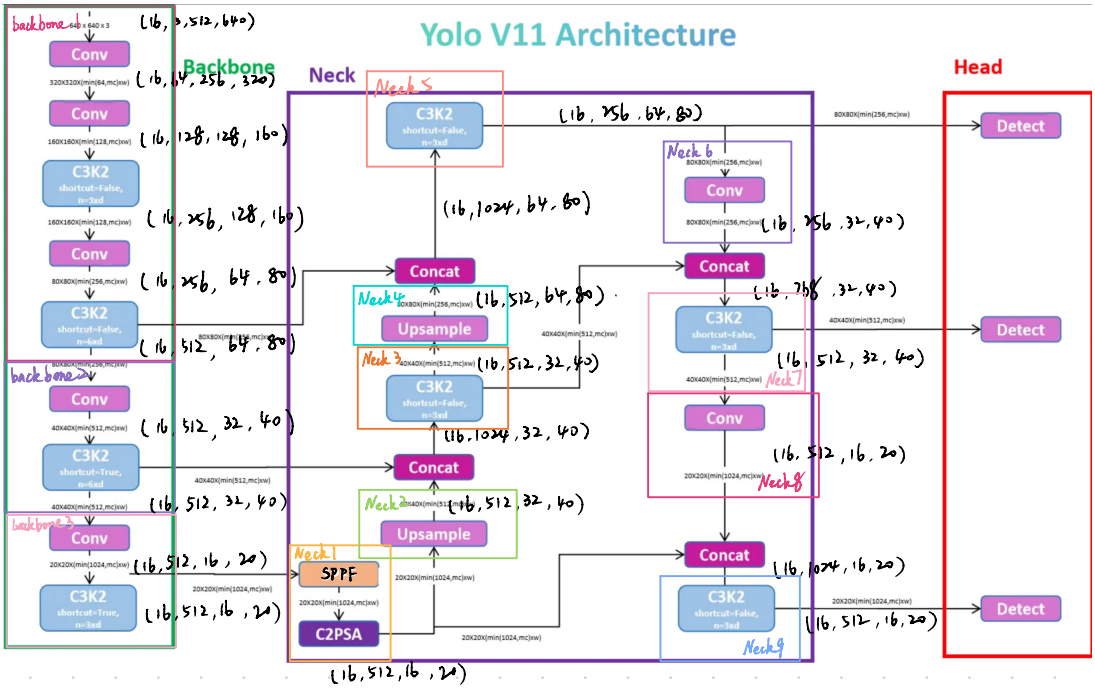
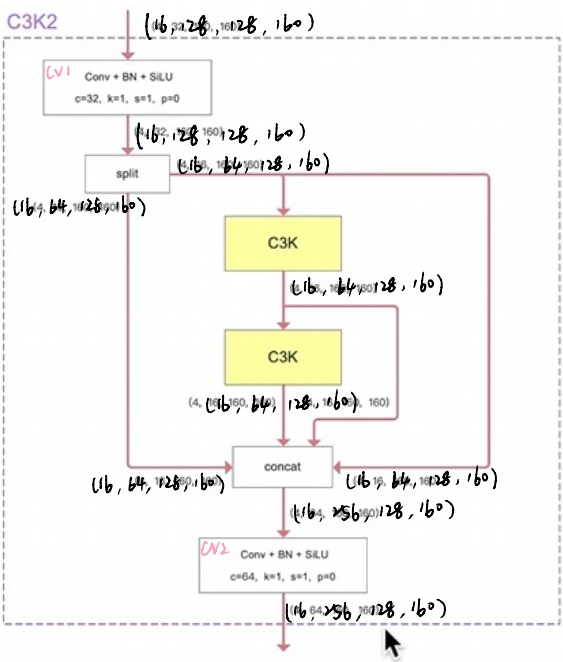
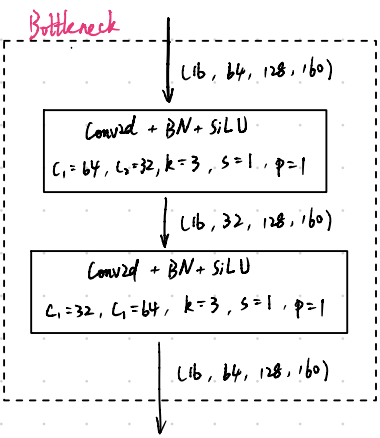
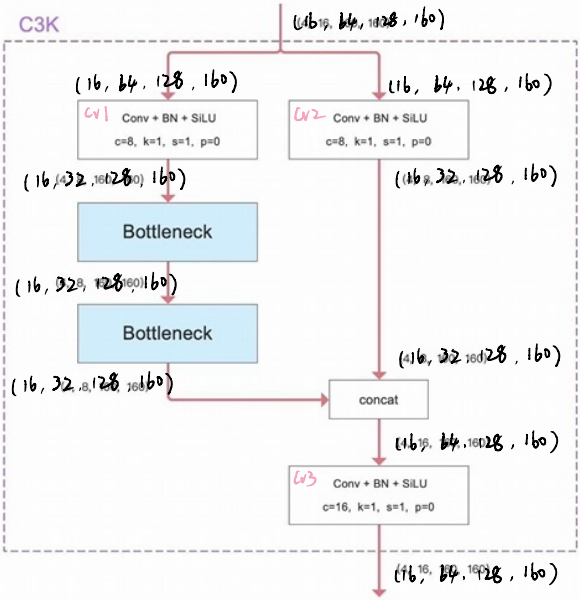
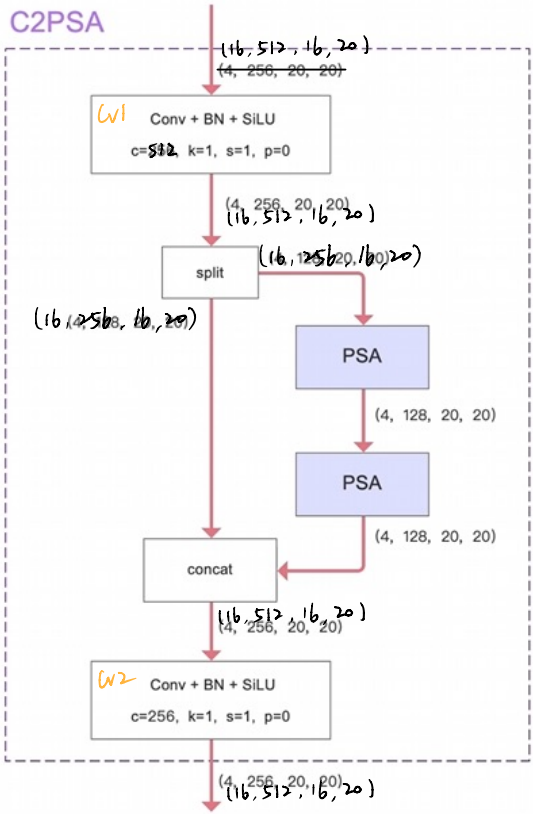
(batch, channel, height, width)

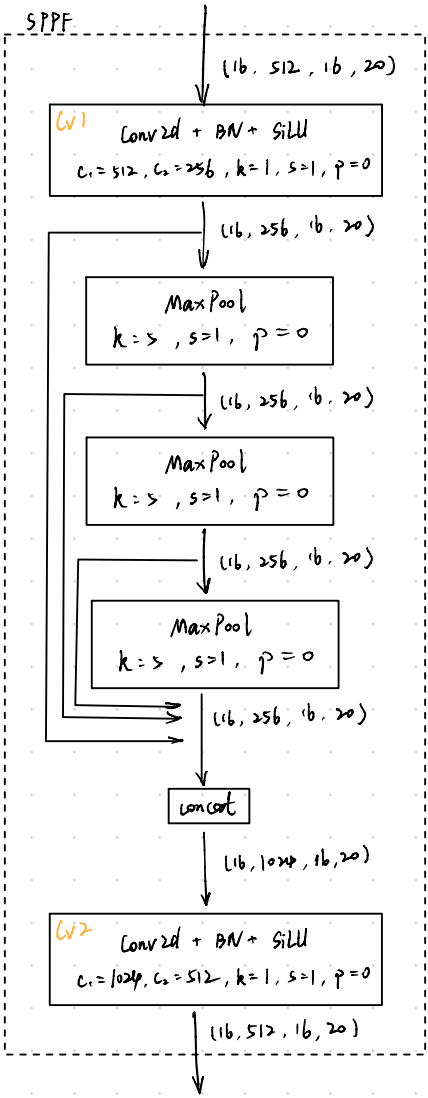


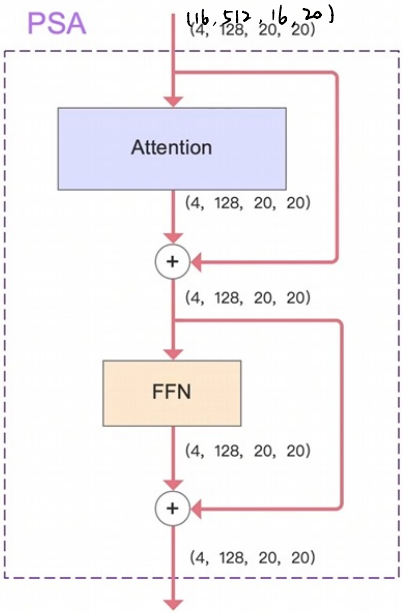


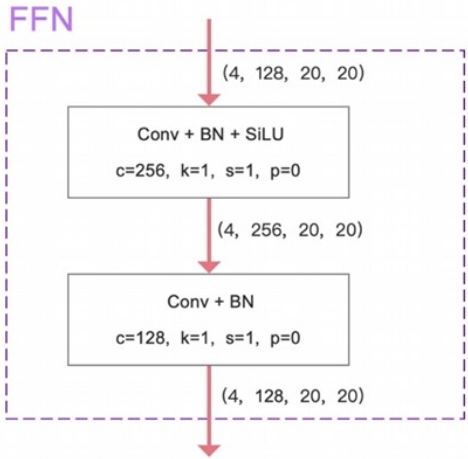


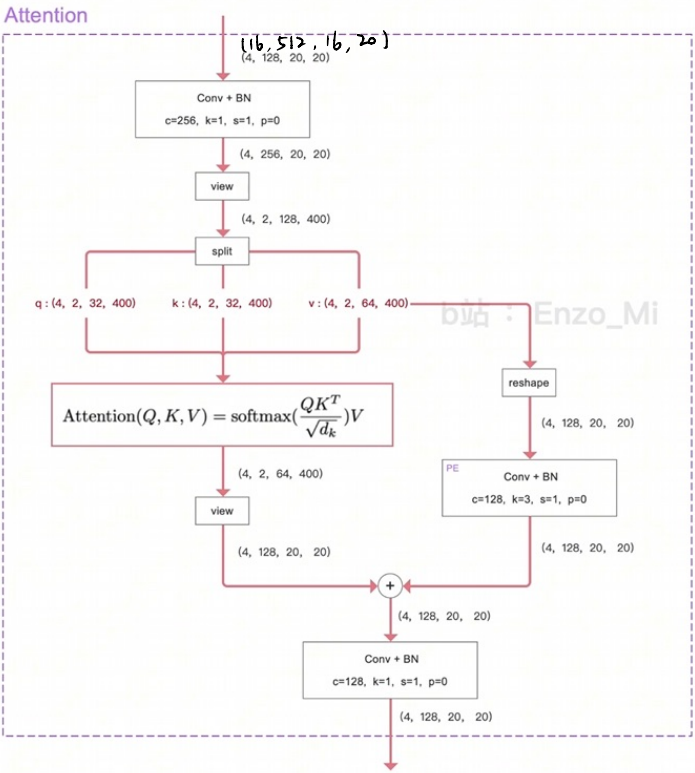

C3K
(16, 64, 128, 160)
(16, 64, 128, 160)
(16, 64, 128, 160)
CV1
Conv + BN + SiLU
c=8, k=1, s=1, p=0
CV2
Conv + BN + SiLU
c=8, k=1, s=1, p=0
(16, 32, 128, 160)
(16, 32, 128, 160)
Bottleneck
(16, 32, 128, 160)
Bottleneck
(16, 32, 128, 160)
(16, 32, 128, 160)
concat
(16, 64, 128, 160)
CV3
Conv + BN + SiLU
c=16, k=1, s=1, p=0
(16, 64, 128, 160)

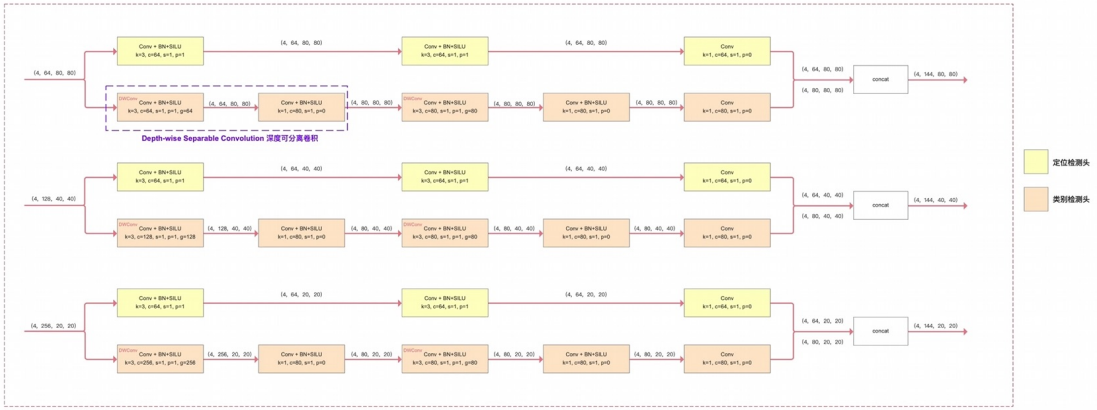

C2PSA
(16, 512, 16, 20)
(4, 256, 20, 20)
CV1
Conv + BN + SiLU
c=512, k=1, s=1, p=0
(4, 256, 20, 20)
(16, 512, 16, 20)
split
(16, 256, 16, 20)
(16, 256, 16, 20)
PSA
(4, 128, 20, 20)
PSA
(4, 128, 20, 20)
concat
(16, 512, 16, 20)
(4, 256, 20, 20)
CV2
Conv + BN + SiLU
c=256, k=1, s=1, p=0
(4, 256, 20, 20)
(16, 512, 16, 20)


SPPF
(16, 512, 16, 20)
CV1
Conv2d + BN + SiLU
$c_1 = 512$, $c_2 = 256$, $k=1$, $s=1$, $p=0$
(16, 256, 16, 20)
MaxPool
$k=5$, $s=1$, $p=0$
(16, 256, 16, 20)
MaxPool
$k=5$, $s=1$, $p=0$
(16, 256, 16, 20)
MaxPool
$k=5$, $s=1$, $p=0$
(16, 256, 16, 20)
concat
(16, 1024, 16, 20)
CV2
Conv2d + BN + SiLU
$c_1 = 1024$, $c_2 = 512$, $k=1$, $s=1$, $p=0$
(16, 512, 16, 20)


PSA
(16, 512, 16, 20)
(4, 128, 20, 20)
Attention
(4, 128, 20, 20)
+
(4, 128, 20, 20)
FFN
(4, 128, 20, 20)
+
(4, 128, 20, 20)


FFN
(4, 128, 20, 20)
Conv + BN + SiLU
c=256, k=1, s=1, p=0
(4, 256, 20, 20)
Conv + BN
c=128, k=1, s=1, p=0
(4, 128, 20, 20)


Attention
(16, 512, 16, 20)
(4, 128, 20, 20)
Conv + BN
c=256, k=1, s=1, p=0
(4, 256, 20, 20)
view
(4, 2, 128, 400)
split
q : (4, 2, 32, 400)
k : (4, 2, 32, 400)
v : (4, 2, 64, 400)
b站 : Enzo_Mi
reshape
(4, 128, 20, 20)
$\mathrm{Attention}(Q,K,V)=\mathrm{softmax}(\frac{QK^T}{\sqrt{d_k}})V$
(4, 2, 64, 400)
PE
Conv + BN
c=128, k=3, s=1, p=0
view
(4, 128, 20, 20)
(4, 128, 20, 20)
+
(4, 128, 20, 20)
Conv + BN
c=128, k=1, s=1, p=0
(4, 128, 20, 20)


(4, 64, 80, 80)
Conv + BN+SiLU
k=3, c=64, s=1, p=1
(4, 64, 80, 80)
Conv + BN+SiLU
k=3, c=64, s=1, p=1
(4, 64, 80, 80)
Conv
k=1, c=64, s=1, p=0
DWConv
Conv + BN+SiLU
k=3, c=64, s=1, p=1, g=64
(4, 64, 80, 80)
Conv + BN+SiLU
k=1, c=80, s=1, p=0
(4, 80, 80, 80)
DWConv
Conv + BN+SiLU
k=3, c=80, s=1, p=1, g=80
(4, 80, 80, 80)
Conv + BN+SiLU
k=1, c=80, s=1, p=0
(4, 80, 80, 80)
Conv
k=1, c=80, s=1, p=0
Depth-wise Separable Convolution 深度可分离卷积
(4, 64, 80, 80)
(4, 80, 80, 80)
concat
(4, 144, 80, 80)
(4, 128, 40, 40)
Conv + BN+SiLU
k=3, c=64, s=1, p=1
(4, 64, 40, 40)
Conv + BN+SiLU
k=3, c=64, s=1, p=1
(4, 64, 40, 40)
Conv
k=1, c=64, s=1, p=0
DWConv
Conv + BN+SiLU
k=3, c=128, s=1, p=1, g=128
(4, 128, 40, 40)
Conv + BN+SiLU
k=1, c=80, s=1, p=0
(4, 80, 40, 40)
DWConv
Conv + BN+SiLU
k=3, c=80, s=1, p=1, g=80
(4, 80, 40, 40)
Conv + BN+SiLU
k=1, c=80, s=1, p=0
(4, 80, 40, 40)
Conv
k=1, c=80, s=1, p=0
(4, 64, 40, 40)
(4, 80, 40, 40)
concat
(4, 144, 40, 40)
(4, 256, 20, 20)
Conv + BN+SiLU
k=3, c=64, s=1, p=1
(4, 64, 20, 20)
Conv + BN+SiLU
k=3, c=64, s=1, p=1
(4, 64, 20, 20)
Conv
k=1, c=64, s=1, p=0
DWConv
Conv + BN+SiLU
k=3, c=256, s=1, p=1, g=256
(4, 256, 20, 20)
Conv + BN+SiLU
k=1, c=80, s=1, p=0
(4, 80, 20, 20)
DWConv
Conv + BN+SiLU
k=3, c=80, s=1, p=1, g=80
(4, 80, 20, 20)
Conv + BN+SiLU
k=1, c=80, s=1, p=0
(4, 80, 20, 20)
Conv
k=1, c=80, s=1, p=0
(4, 64, 20, 20)
(4, 80, 20, 20)
concat
(4, 144, 20, 20)
定位检测头
类别检测头